

Generative Modelling for Realistic Mars Landing Site Visualisation

Interim Draft

Deepakraj Rajmohan

Master of Science in Artificial Intelligence

from the

University of Surrey



School of Computer Science and Electrical and Electronic Engineering

Faculty of Engineering and Physical Sciences

University of Surrey

Guildford, Surrey, GU2 7XH, UK

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Supervised by: Dr Xiatian Zhu

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“Generative Modelling for Realistic Mars Landing Site Visualisation”

Deepakraj Rajmohan

Author Signature

Date:

Supervisor's name: Dr Xiatian Zhu

WORD COUNT

This is an interim draft covering the chapters completed to date (Introduction through Results and Diagnosis). A Conclusions chapter has not yet been written, so the counts below will change substantially before final submission.

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ABSTRACT

Mission planning and astronaut/operator training for planetary exploration increasingly rely on immersive VR reconstructions of candidate landing sites, but the only data available for most such sites is nadir orbital imagery (e.g. NASA’s HiRISE, 25 cm/px), not the ground-level, perspective views that make a VR scene feel navigable. This dissertation addresses that nadir-to-perspective domain gap in the context of ESA’s ExoMars Rosalind Franklin rover mission, proposing a three-stage generative pipeline — diffusion-based super-resolution, physics-constrained unpaired view translation, and Gaussian Splatting scene reconstruction — to turn orbital imagery into explorable, physically plausible VR environments.

To date, the literature review and methodology have been completed, surveying the super-resolution, image-to-image translation, neural rendering, and hallucination-safety literature relevant to each stage, and formalising the proposed Neural Schrödinger Bridge formulation, evaluation framework (FID, shadow-angle error, hallucination rate, VR frame rate), and baseline comparisons (MARSGAN, DiffusionSat, CycleGAN, UNSB-no-physics). A CycleGAN baseline for Stage 2 has been implemented, trained, and evaluated end-to-end: a real-resolution HiRISE data pipeline was built after an initial corpus was found to use browse-quality (~ 3 m/px) imagery rather than the true 25 cm/px RDR products, and a corrected $\sim 30,000$ -patch corpus was assembled from seven Oxia Planum observations. A 50-epoch training run on this corrected corpus was completed and evaluated (FID 269.9), and a specific, evidenced diagnosis was reached: one discriminator collapses from epoch 3 onward, most likely because the grainy, true-resolution HiRISE patches hand it an easy non-semantic shortcut over the smoother rover imagery, starving the corresponding generator of a useful gradient. In parallel, a CTX/HiRISE co-registration pipeline for the Stage 1 super-resolution corpus is under development.

This draft covers the work completed to date; addressing the CycleGAN discriminator imbalance, building the physics-constrained UNSB stage, and evaluating the full pipeline remain the subject of ongoing work.

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1 INTRODUCTION

Planetary exploration missions increasingly depend on immersive virtual-reality (VR) environments for mission planning, hazard assessment, and astronaut or rover-operator training. Building a convincing VR reconstruction of a Mars landing site, however, runs into a basic data problem: the only imagery available for almost every candidate site, including all ExoMars candidate landing areas, is nadir orbital imagery captured from orbit looking straight down, whereas a usable VR scene needs ground-level, perspective imagery of the kind a rover’s own cameras produce after it has landed. This dissertation is concerned with closing that nadir-to-perspective domain gap using generative modelling, so that orbital data alone — available years before any rover touches down — can be turned into a physically plausible, explorable VR surface.

The work is computational and experimental in nature: it combines remote-sensing data engineering (acquiring and correctly processing real orbital and rover imagery), generative model design and training (image-to-image translation and super-resolution networks), and, in later stages, real-time neural rendering. The concrete application context is ESA’s ExoMars Rosalind Franklin rover mission, and the primary test site is Oxia Planum, the mission’s selected landing area.

1.1 Background and Context

High-resolution orbital imagery of Mars is comparatively abundant: NASA’s HiRISE instrument has imaged large fractions of the surface at up to 25 cm/pixel, resolving individual boulders and small craters. This is nadir imagery — captured looking straight down from orbit — and it is systematically unlike the perspective, oblique, ground-level imagery captured by rover cameras such as NAVCAM, which show terrain from roughly a metre above the surface, with strong foreshortening, occlusion, and local lighting effects that a top-down orbital image simply does not contain. No dataset of geographically *aligned* nadir/perspective Mars image pairs exists, because rover imagery only exists for the handful of sites a rover has actually visited, and even then the rover was never positioned to recreate an orbital viewing geometry. Any method that aims to synthesise rover-perspective imagery from orbital data must therefore work without paired supervision.

This gap sits at the intersection of several active research areas surveyed in Chapter 2: single-image super-resolution (to recover resolution lost between coarser orbital instruments and HiRISE),

unpaired image-to-image translation (CycleGAN-style methods, and more recent physics-informed variants such as the Neural Schrödinger Bridge), neural scene rendering (NeRF and Gaussian Splatting for turning a set of synthetic views into an explorable 3D scene), and the emerging concern, specific to generative models used for scientific and mission-critical purposes, of hallucination: a model may produce a plausible-looking rock or crater that was never actually present, which is a materially different failure mode from a blurry or low-fidelity output. Existing single-modality approaches (pure super-resolution, or pure style-transfer GANs) each solve part of this problem but none combines resolution recovery, physically-grounded view translation, and real-time rendering into a single pipeline validated against Mars-specific physical plausibility criteria, which is the gap this dissertation addresses.

1.2 Objectives

The objectives below were carried forward from the initial project proposal and refined during the methodology stage (Chapter 3) once the literature review had established that no existing method closes the nadir-to-rover-perspective gap end-to-end for Mars surface synthesis:

- **O1.** Produce nadir imagery at 25 cm/pixel from coarser CaSSIS/CTX inputs when HiRISE coverage is absent.
- **O2.** Translate unpaired HiRISE nadir patches into statistically plausible rover-perspective images whose photometric and morphometric properties are consistent with Mars surface physics.
- **O3.** Reconstruct a real-time renderable 3D scene from the synthetic perspective images at VR frame rates (≥ 72 fps).
- **O4.** Quantify the fidelity and physical plausibility of generated imagery against held-out real rover data and known planetary surface statistics.

Work to date has focused on the data engineering and baseline modelling required to make O1 and O2 tractable: building correct, real-resolution training corpora for both the orbital and rover domains, and implementing, training, and rigorously evaluating a CycleGAN baseline for O2 before moving to the physics-constrained architecture the methodology proposes. O3 and O4 remain future work.

1.3 Achievements

To date, the following has been completed:

- A full literature review (Chapter 2) covering seven topic clusters — orbital data constraints, super-resolution, image-to-image translation, digital terrain modelling, neural scene rendering, VR reconstruction, and generative hallucination — synthesising 20 papers into a comparative analysis that motivates the proposed architecture.
- A complete methodology (Chapter 3) formalising the three-stage pipeline, including the Neural Schrödinger Bridge formulation with four physics-informed loss terms, and a full evaluation protocol against four baselines.
- A real-resolution data pipeline (Chapter 4) that corrected a data-quality bug in the original HiRISE corpus (browse-quality ~3 m/px imagery masquerading as the intended 25 cm/px product) and rebuilt it from genuine RDR products under a tight disk-quota constraint on the training machine.
- A trained and evaluated CycleGAN baseline (Chapter 5) for the Stage 2 view-translation problem, including a specific, evidence-based diagnosis of why its current FID score (269.9) falls well short of the target (≤ 55): a discriminator collapse traced to the epoch level using the training logs, not a vague appeal to “GANs are unstable”.
- An in-progress CTX/HiRISE co-registration pipeline for the Stage 1 super-resolution corpus, with a working smoke test on five image pairs.

1.4 Overview of Dissertation

Chapter 2 reviews the literature underpinning each stage of the proposed pipeline and identifies the specific gap this work fills. Chapter 3 presents the methodology: the mathematical formulation of the three-stage architecture, the dataset design, and the evaluation framework. Chapter 4 describes what has actually been implemented so far — the data acquisition and preprocessing pipelines for both the HiRISE and rover corpora, including the real-resolution corpus rebuild, and the CycleGAN model and training infrastructure built to establish a Stage 2 baseline. Chapter 5 presents the results of training that baseline: the loss behaviour over 50 epochs, the FID evaluation, representative sample translations, and a diagnosis of the specific failure mode observed,

together with candidate fixes. Resolving that diagnosis, building the physics-constrained Neural Schrödinger Bridge, completing the Stage 1 super-resolution corpus, and implementing and evaluating the Gaussian Splatting rendering stage remain as future work. This draft does not yet include a final Results/Evaluation chapter for the completed system or a Conclusions chapter, since that work is still in progress.

2 BACKGROUND THEORY AND LITERATURE REVIEW

This chapter surveys the literature underpinning the proposed pipeline and identifies the specific gap the dissertation addresses. It is condensed from a standalone literature review produced earlier in the project; the full nine-page version, including additional discussion of each publication, is available alongside this dissertation. The chapter is organised around seven thematic clusters — orbital data characterisation, super-resolution, image-to-image translation, digital terrain modelling, neural scene rendering, VR reconstruction, and generative hallucination — before a comparative synthesis (Section 2.6) draws out the cross-cutting patterns that motivate the proposed architecture.

2.1 Taxonomy and Conceptual Framework

The literature reviewed here spans several independently advancing communities — remote sensing image processing, generative modelling, planetary science, and VR engineering — whose outputs are jointly necessary to the downstream synthesis requirement without any of them addressing it explicitly. Three axes organise the territory: *spatial view* (nadir/orbital vs. perspective/rover vs. both), *task type* (enhancement, translation, or geometric reconstruction), and *output modality* (RGB texture, elevation/depth, or rendered video). Seven clusters emerge: orbital data characterisation [16, 13]; nadir super-resolution [17, 19, 11, 8, 21]; image-to-image translation [9, 20]; monocular digital terrain modelling [18, 22, 2]; neural scene rendering [5, 10, 4, 12, 14]; VR reconstruction [3, 10]; and generative reliability [1, 15].

The gap between the super-resolution cluster and the neural-rendering cluster is the primary concern of this dissertation. Nadir SR methods keep the viewpoint fixed; neural rendering methods require perspective imagery as input. No reviewed method translates from nadir to perspective in a way that synthesises rover-viewable texture from orbital data — a structural gap reflecting the distinct communities that have developed these tools, not an oversight.

2.2 Super-Resolution for Planetary Remote Sensing

Super-resolution research on planetary imagery has progressed through three generations since 2021. Tao *et al.* [17] propose MARSGAN, a CaSSIS-to-HiRISE-equivalent enhancement of ESRGAN achieving roughly $3\times$ effective resolution improvement, evaluated only qualitatively across

eight Mars sites — co-registering CaSSIS and HiRISE precisely enough for pixel-level comparison remains non-trivial, so quantitative cross-method comparison stays opaque throughout this sub-field. Wu *et al.* [19] apply recursive Swin Transformer attention (RSTSRN) at $\times 2$ – $\times 8$, improving on LapSRN by up to 1.22 dB PSNR at $\times 8$, but evaluate only against bicubically down-sampled inputs rather than the genuinely distinct point-spread function of a real orbital sensor — a protocol weakness that recurs across this literature and directly informed how the corpus for this dissertation’s own CycleGAN baseline was later validated (Chapter 4).

Khanna *et al.* [8] propose DiffusionSat, a diffusion foundation model for satellite imagery conditioned on geolocation and sensor metadata, demonstrating that multi-sensor domain generalisation is tractable within a diffusion framework, though its training corpus contains no Mars data and its conditioning assumes Earth coordinates. Zhu *et al.* [21] address the fidelity/perceptual-realism trade-off directly relevant to preserving large-scale geomorphology while generating sub-metre texture, via frequency-decomposition conditioning that anchors denoising to low-frequency structure. Lu and Su [11] fuse THEMIS thermal imagery with visible and topographic data for SR, of indirect relevance since thermal is not the visible RGB domain required here.

Collectively, this literature shows $\times 4$ enhancement is achievable with GAN, transformer, or diffusion backbones, but every reviewed method operates nadir-to-nadir: input and output share a viewpoint. The perspective transformation this dissertation requires is nowhere addressed by SR alone.

2.3 Image-to-Image Translation as a Domain-Bridging Mechanism

Image-to-image translation methods map entire image domains to each other, structurally better suited to a view transformation than SR. Kim *et al.* [9] propose UNSB (Unpaired Neural Schrödinger Bridge), reframing unpaired translation as learning an SDE that transports directly between two target distributions rather than routing through a Gaussian prior, decomposed into a sequence of adversarial sub-problems for tractability at high resolution. UNSB outperforms CycleGAN and CUT on standard unpaired benchmarks (horse-to-zebra, summer-to-winter) and is architecturally compelling here precisely because it requires no geographic co-registration between domains — but it has only been evaluated on natural-image colour-domain shifts, far milder than the shadow-geometry, foreshortening, and occlusion differences between a nadir and a perspective view of the same Mars terrain. Xiao *et al.* [20] address the complementary reproducibility problem — stochastic I2I methods yield different outputs on successive runs, unacceptable for

scientific visualisation — via a Dual-approx Bridge with separate forward/reverse approximators, achieving deterministic translation with superior SSIM/LPIPS versus stochastic baselines, though again untested on remote sensing imagery.

The methodology (Chapter 3) builds on UNSB as the target Stage 2 architecture, with a Cycle-GAN baseline (Chapter 4–5) implemented first precisely because, as this section establishes, no reviewed I2I method has been validated against a domain gap this large, making an empirically-grounded baseline a necessary first step rather than a formality.

2.4 Digital Terrain Modelling and Neural Scene Rendering

Monocular DTM recovery is a necessary precursor to perspective-view synthesis but does not itself produce visual texture. Tao *et al.* [18] recover pixel-scale topography from single HiRISE images via photoclinometry (MADNet 2.0); Zou *et al.* [22] improve on this with PFMGAN, an explicit solar-geometry and albedo-aware GAN that reduces reconstruction RMSE by over 50% across five landform types — the strongest quantitative evidence in the survey that physically-grounded architectures outperform unconstrained ones, and a direct design precedent for the physics-informed loss terms in the proposed Stage 2 methodology. Cao *et al.* [2] map HiRISE orthoimages to dense DEMs via conditional GAN. None of these three produce rendered surface appearance.

Neural rendering methods are architecturally capable of view-angle transformation but require multi-view input imagery of the *same* scene, which does not exist for a not-yet-visited landing site. Giusti *et al.* [5] demonstrate NeRF converges on Curiosity rover imagery (MaRF); Li *et al.* [10] build a complete text-conditioned video-diffusion simulation pipeline (MarsGen) from over 10,000 annotated Curiosity clips, the most operationally complete system reviewed, but entirely dependent on rover imagery that only exists for four previously-visited sites. Martínez-Petersen *et al.* [12] couple NeRF and Gaussian Splatting in a ROS 2 pipeline, establishing that Gaussian Splatting’s interactive render rate is the correct representation target for the VR output this dissertation aims at (Stage 3), even though their pipeline also requires rover imagery as input. Paul *et al.* [14]’s survey confirms, as its central negative finding, that synthesising a NeRF-equivalent scene from a single nadir orbital image has not been addressed anywhere in the literature.

2.5 Virtual Reality Reconstruction and Generative Reliability

Catalano and Soccini [3] inpaint void regions in Mars heightmaps via a boundary-conditioned diffusion model, improving RMSE by up to 15% and LPIPS by up to 81% over classical interpolation — but a completed heightmap carries no colour information, so this addresses only the geometric half of the VR terrain problem, leaving texture synthesis explicitly open.

Every generative model surveyed can, in principle, produce plausible surface features with no basis in physical reality — an aesthetic non-issue for natural images but a safety-relevant failure mode for mission-critical planetary visualisation. Aithal *et al.* [1] identify *mode interpolation* as the mechanism: a diffusion model’s smooth score-function approximation assigns non-zero probability to samples interpolated between nearby training modes (e.g. two crater morphologies) that never occurred in training data, a structural property of the approximation, not a defect correctable with more training. Rathkopf [15] reframes this as an epistemological question, arguing from AlphaFold 3 and GenCast case studies that hallucination risk becomes bounded and scientifically acceptable only through theory-guided training (domain physical constraints) combined with confidence-based error screening. Together, these establish that physical constraint is a primary mechanism for bounding hallucination risk, not an optional refinement — directly motivating the physics-informed loss terms specified in the methodology (Section 3.2.3) rather than treating Stage 2 as an unconstrained image-translation problem.

2.6 Comparative Synthesis

Table 2.1 consolidates the twenty reviewed publications. A clean partition separates nadir-domain methods (all SR and DTM papers) from perspective-domain methods (all NeRF papers evaluated on real Mars data) — no method crosses this boundary except the two I2I papers, benchmarked only on natural images. The methods with the strongest quantitative improvement (PFMGAN, over 50% RMSE reduction) are those with explicit physical constraints, while unconstrained SR methods show greater susceptibility to the mode-interpolation artefacts Aithal *et al.* describe. The two systems closest to a complete VR pipeline — MarsGen and the heightmap-inpainting system — each solve exactly one dimension (perspective texture *or* elevation, never both from nadir input). No reviewed system combines orbital nadir input, perspective-view RGB texture output, and physical-plausibility constraints.

Table 2.1: Comparative overview of the reviewed literature against the nadir-to-rover-perspective VR synthesis problem. Task codes: SR = super-resolution; I2I = image-to-image translation; DTM = digital terrain model generation; NeRF/GS = neural radiance field / Gaussian splatting; VR = VR reconstruction; Data = dataset; Theory = theoretical analysis; Review = survey. View codes: N = nadir/orbital; P = perspective/rover; 3D = 3D representation.

Ref.	Method	Task	View	Key Result	Mars?
[17]	MARSGAN	SR	N→N	$\approx 3\times$ SR on CaSSIS, qualitative only	Yes
[19]	RSTSRN	SR	N→N	+0.88 dB PSNR vs. LapSRN ($\times 4$)	Yes
[11]	Multi-source Thermal SR	SR	N→N	PSNR gain on THEMIS thermal	Yes
[8]	DiffusionSat	SR/Gen	N→N	SOTA FID, multi-sensor	No
[21]	TamingDiff-SR	SR	N→N	$\times 4$, PSNR + competitive FID	No
[9]	UNSB	I2I	Any	Lower FID vs. CycleGAN/CUT, unpaired	No
[20]	Dual-approx Bridge	I2I	Any	Deterministic I2I, higher SSIM/LPIPS	No
[18]	MADNet 2.0	DTM	N→3D	Monocular DTM $\approx$ stereo accuracy	Yes
[22]	PFMGAN	DTM	N→3D	>50% RMSE reduction, 5 landform types	Yes
[2]	Cond. GAN-DTM	DTM	N→3D	Height consistency, 4 Mars regions	Yes
[13]	MCTED dataset	Data	N (pairs)	80,898 CTX–DEM pairs	Yes
[16]	Bayesian rock density	Analysis	N→N	Probabilistic density from CTX	Yes
[5]	MaRF	NeRF	P→3D	Novel view synthesis $\approx$ standard NeRF	Yes
[10]	MarsGen	NeRF+VR	P→VR	Beats terrestrial video models on FID	Yes
[4]	Neural height-field	NeRF	N/oblique	Beats MVS on RMSE (simulated descent)	Partial
[12]	NeRF+GS pipeline	NeRF+GS	P→3D	Real-time GS render, ROS 2 integration	Partial
[14]	NeRF-in-space survey	Review	—	Confirms nadir→perspective gap is open	Yes
[3]	Diffusion inpainting	VR	N→VR _H	15% RMSE / 81% LPIPS improvement	Yes
[1]	Mode interpolation	Theory	—	Mechanistic account of hallucination	No
[15]	Hallucination-to-reliability	Theory	—	Physics-guided training bounds risk	No

2.7 Summary

The literature establishes that each constituent capability needed for VR-grade Mars landing-site visualisation — orbital super-resolution, unpaired image-to-image translation, terrain reconstruction, neural rendering, and hallucination-aware training — has individually advanced substan-

tially since 2021, but no reviewed method integrates them into a pipeline that starts from nadir orbital data and ends with physically-plausible, rover-perspective visual texture. This gap, and the specific finding that physical constraints (not perceptual loss alone) are what separates the strongest-performing methods in each cluster from the weakest, directly motivate the three-stage, physics-constrained architecture formalised in Chapter 3.

3 METHODOLOGY

This chapter formalises the three-stage physics-constrained generative pipeline motivated in Chapter 2: a mathematical problem formulation, the proposed system architecture, dataset design, training strategy, and evaluation framework, condensed from a standalone methodology document produced earlier in the project.

3.1 Problem Formulation

Let $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ denote an image tensor. Define two domains: the nadir orbital domain $\mathcal{X} = \{\mathbf{x}_i^{\text{nad}}\}_{i=1}^N$ comprising HiRISE patches at 25 cm/pixel, and the rover-perspective domain $\mathcal{Y} = \{\mathbf{y}_j^{\text{rov}}\}_{j=1}^M$ comprising NAVCAM images, with no correspondence between index sets i and j — the training regime is fully unpaired, with respective marginal distributions P_{nad} and P_{persp} .

Stage 1 (optional, super-resolution). Given a low-resolution CaSSIS/CTX image at 4–6 m/pixel, Stage 1 produces a HiRISE-equivalent estimate at 0.25 m/pixel (16–24 $\times$), following the conditional diffusion framework of DiffusionSat [8] and a taming-diffusion SR approach [21]:

$$d\mathbf{z}_t = [f(\mathbf{z}_t, t) - g(t)^2 s_\theta(\mathbf{z}_t, t, \mathbf{x}^{\text{lr}})] dt + g(t) d\bar{\mathbf{W}}_t, \quad (3.1)$$

where s_θ is a denoising score function conditioned on the low-resolution input. This stage is bypassed when HiRISE imagery is already available directly, which is the case for the primary test site, Oxia Planum.

Stage 2 (core contribution, view translation). The Neural Schrödinger Bridge (UNSB) [9] solves the entropy-regularised optimal transport problem between P_{nad} and P_{persp} :

$$\pi^* = \arg \min_{\pi \in \Pi(P_{\text{nad}}, P_{\text{persp}})} \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \pi} [c(\mathbf{x}, \mathbf{y})] + \varepsilon \text{KL}(\pi \parallel R), \quad (3.2)$$

realised by a forward–backward SDE pair jointly trained via the Iterative Proportional Fitting Procedure (IPFP), with a Brownian Bridge ODE variant [20] available where deterministic outputs are required.

Because unconstrained translation networks trained on Mars data are prone to photometric hallucination [1, 15], four physics-informed loss terms are appended to the base bridge loss $\mathcal{L}_{\text{SB}}$: a solar-geometry consistency term $\mathcal{L}_{\text{solar}}$ penalising deviation between the generated image’s shadow direction and the direction implied by the HiRISE acquisition’s sun angle metadata; a Hapke photometric term $\mathcal{L}_{\text{photo}}$ constraining the estimated single-scattering albedo to the Mars ferric-oxide prior via the Hapke BRDF [6]; a geological morphometry term $\mathcal{L}_{\text{geo}}$ scoring generated patches against known crater size-frequency and aeolian bedform statistics; and a standard VGG-based perceptual term $\mathcal{L}_{\text{perceptual}}$. The total objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SB}} + \lambda_1 \mathcal{L}_{\text{solar}} + \lambda_2 \mathcal{L}_{\text{photo}} + \lambda_3 \mathcal{L}_{\text{geo}} + \lambda_4 \mathcal{L}_{\text{perceptual}}, \quad (3.3)$$

with initial weights $\lambda_1 = 0.5$, $\lambda_2 = 1.0$, $\lambda_3 = 0.8$, $\lambda_4 = 0.1$, set from a preliminary scale analysis and refined by validation-set search.

Stage 3 (rendering). Given K synthetic perspective images with estimated camera poses, a 3D Gaussian Splatting scene [7] is fit by minimising a photometric reconstruction loss combining ℓ_1 and D-SSIM terms over the differentially-rasterised views, producing a scene renderable in real time for VR display.

3.2 Proposed System Architecture

The three stages are designed to be modular: Stage 2 and Stage 3 can operate independently of Stage 1 whenever HiRISE coverage is already available, which is the case for the primary Oxia Planum test site. Stage 2’s generator G_ϕ is a 256-channel U-Net with eight residual blocks and self-attention at 32×32 and 16×16 scales; solar azimuth/elevation metadata are injected at each residual block via adaptive layer normalisation, following the conditioning strategy of PFM-GAN [22]. UNSB is chosen over CycleGAN and standard diffusion inpainting for three reasons: its entropy-regularised optimal-transport objective gives a principled probabilistic account of the domain gap; it supports fully unpaired training, essential since no geographically registered nadir–perspective Mars pairs exist; and its SDE formulation supports switching to a deterministic Brownian Bridge ODE when repeatable outputs are required for evaluation or multi-view consistency. Stage 3 uses the reference 3DGS implementation [7] unmodified, seeded from MADNet-2.0 depth estimates [18] to reduce floater artefacts on the textureless regions common in Mars terrain.

A CycleGAN baseline (unpaired, no physics constraints) is implemented first, ahead of the full UNSB, precisely because Chapter 2 established that no reviewed I2I method has been validated against a domain gap this large — an empirically-grounded, well-understood baseline is a necessary first step, not a formality. Its implementation, training, and diagnosis are reported in Chapters 4–5.

3.3 Dataset Design and Acquisition Strategy

The primary geographic focus is Oxia Planum (18.3°N, 335.5°E), the ExoMars 2028 candidate landing site, imaged by HiRISE on more than 30 occasions. HiRISE products are radiometrically corrected (RDR) and tiled at 25 cm/pixel; tiles with cloud or data-gap fractions exceeding 5% are discarded. Rover NAVCAM imagery is drawn from the Curiosity and Perseverance raw image archives, filtered to remove dust-storm-opacity acquisitions and frames showing rover hardware. Table 3.1 summarises the target corpus. Both corpora are split 80/10/10 (train/val/test), stratified by geographic region to avoid spatial-autocorrelation leakage; augmentation (flips, 90° rotations, brightness jitter) is applied only to training data, with solar-metadata angles transformed consistently with any spatial flip.

Table 3.1: Target dataset summary (methodology design target; the corpus actually built to date is reported in Chapter 4).

Corpus	Size	Resolution	Source
HiRISE nadir patches	50,000	25 cm/px	NASA PDS HiRISE RDR
Rover NAVCAM images	30,000	≤ 1280 px	NASA raw archives
Total	80,000	—	—

3.4 Training Strategy and Optimisation

Stage 1 is designed to be initialised from a DiffusionSat checkpoint [8] and fine-tuned on Mars orbital data. Stage 2 is designed around the IPFP procedure of [9]: alternating forward/backward half-bridge updates for 10 outer iterations, with three auxiliary loss networks (a shadow-direction estimator, a Hapke inversion network, and a geological in/out-of-distribution classifier) pre-trained separately and kept frozen during UNSB training. Stage 3 trains the 3DGS scene for approximately 30,000 iterations with periodic densification and opacity pruning, following [7]. Algorithm 3.1 summarises the Stage 2 training loop as designed; its implementation has not yet begun.

Algorithm 1: Stage 2 UNSB Training with Physics Losses

Input: HiRISE corpus $\mathcal{X}$; rover corpus $\mathcal{Y}$; pretrained auxiliary nets D_ψ , Ω_ξ , K_κ ; loss weights $\lambda_{1..4}$; IPFP iterations $N_{\text{ipfp}} = 10$

Output: Trained generator G_ϕ

```

for  $n = 1$  to  $N_{\text{ipfp}}$  do
  // Forward half-bridge: update  $G_\phi$ 
  for  $t = 1$  to 5000 do
    Sample  $\mathbf{x} \sim \mathcal{X}$ ,  $\mathbf{y} \sim \mathcal{Y}$ ;  $\hat{\mathbf{y}} \leftarrow G_\phi(\mathbf{x})$ 
    Compute  $\mathcal{L}_{\text{SB}}$ ,  $\mathcal{L}_{\text{solar}}$ ,  $\mathcal{L}_{\text{photo}}$ ,  $\mathcal{L}_{\text{geo}}$ ,  $\mathcal{L}_{\text{perceptual}}$ 
     $\mathcal{L} \leftarrow \mathcal{L}_{\text{total}}$  (Eq. 3.3); update  $G_\phi$  via AdamW
  end for
  // Backward half-bridge update (IPFP step)
end for
return  $G_\phi$ 

```

Figure 3.1: Pseudocode for the designed Stage 2 UNSB training loop (not yet implemented).

3.5 Evaluation Framework

Perceptual quality is measured via Fréchet Inception Distance (FID) and LPIPS against held-out real rover imagery. Where high-quality DTMs are available, structural fidelity (SSIM, LPIPS) is measured against synthetic ground-truth perspective renders. Physical plausibility is measured via shadow-angle error (mean absolute angular deviation between predicted and true solar-derived shadow direction) and Hapke BRDF residual (deviation of estimated albedo from the Mars prior). Hallucination rate is defined as the fraction of generated patches the geological classifier flags as out-of-distribution. VR rendering quality is measured by render throughput (fps) and novel-view PSNR/SSIM.

Table 3.2 sets out the four baselines the proposed method is compared against, with the CycleGAN row implemented and evaluated in Chapters 4–5 of this dissertation.

Table 3.2: Proposed method against baseline methods.

Method	View transformation	Physics constraints	Output	Est. FID↓
MARSGAN [17]	Nadir SR only	None	2D image	~180
DiffusionSat (Mars fine-tune) [8]	Nadir SR/inpainting	None	2D image	~140
CycleGAN (naïve)	Nadir-to-perspective	None	2D image	~110
UNSB (no physics loss) [9]	Nadir-to-perspective	None	2D image	~80
Proposed (Ours)	Nadir-to-perspective + 3D VR	Solar, Hapke, geological, perceptual	3DGS scene	≤55 (target)

The literature-derived ~ 110 FID estimate for a naïve CycleGAN baseline is a useful point of comparison for the actual CycleGAN results reported in Chapter 5: the trained baseline currently scores substantially worse (269.9), and the diagnosis in Chapter 5 explains why that comparison is informative rather than merely disappointing.

3.6 Summary

This chapter has formalised the three-stage pipeline’s mathematical basis, architecture, dataset targets, training strategy, and evaluation protocol, including four physics-informed loss terms designed to bound hallucination risk (Chapter 2) rather than relying on perceptual realism alone. The pipeline targets $\text{FID} \leq 55$, shadow-angle error $< 10^\circ$, hallucination rate $< 5\%$, and VR rendering > 72 fps. The remaining chapters report what has actually been built and evaluated against this design so far: the real-resolution data pipeline (Chapter 4) and the CycleGAN baseline for Stage 2 (Chapters 4–5). Implementing the physics-constrained UNSB, Stage 1, and Stage 3 remains future work.

4 IMPLEMENTATION: DATA PIPELINES AND THE CYCLEGAN BASELINE

This chapter describes what has actually been built so far towards the Stage 2 baseline set out in Chapter 3: the development infrastructure, the data acquisition and preprocessing pipelines for both domains, a data-quality bug discovered in the initial HiRISE corpus and the pipeline rebuilt to fix it, and the CycleGAN model and training infrastructure used to establish the baseline evaluated in Chapter 5.

4.1 Development Infrastructure

Model training requires a CUDA-capable GPU that the primary development machine does not have, so a dual-machine workflow was set up: development and testing happen locally, while training runs on a departmental A4000 GPU workstation (`otter70.eps.surrey.ac.uk`). Code moves between the two machines via a private GitHub repository over SSH; the training machine holds a read-only deploy key so it can pull code but never push, matching its role as a training box rather than a development environment. Large data — raw downloads, processed patches, CTX tiles, and model checkpoints — is deliberately excluded from version control, since GitHub’s free tier cannot sensibly hold multi-gigabyte binaries; each machine instead regenerates its own copy of the data via the download/preprocessing scripts described below. Small, genuinely useful provenance records — the training log and FID evaluation results — are the tracked exception, alongside periodic training-progress sample images, so that training runs remain auditable from the repository history alone without needing to re-run them.

4.2 Initial Data Pipeline and CycleGAN Implementation

The initial HiRISE and rover corpora were built by downloading HiRISE browse-quality JPEGs and rover NAVCAM raw images from the NASA Planetary Data System, then preprocessing both into normalised 256×256 patches: a 1st/99th-percentile intensity stretch, an entropy-based filter (discarding low-texture patches below a fixed entropy threshold, since blank sky or featureless terrain patches contribute nothing useful to training) via `preprocess.py`, and an 80/10/10 train/val/test split. This produced an initial corpus of 37,054/4,631/4,633 HiRISE patches and 20,417/2,552/2,553 rover patches.

The CycleGAN model (`cyclegan.py`, `train_cyclegan.py`) follows the standard architecture of Zhu *et al.*: two ResNet-9 generators (11.36M parameters each, encoder–9 residual-blocks–decoder with instance normalisation and reflection padding) and two PatchGAN-70 discriminators (2.76M parameters each, outputting a 30×30 patch-level realism map), trained with LSGAN loss, cycle-consistency weight $\lambda_{\text{cyc}} = 10$, identity-loss weight $\lambda_{\text{idt}} = 5$, Adam ($\text{lr} = 2\times 10^{-4}$, $\beta_1 = 0.5$), a 50-image replay buffer per discriminator to reduce oscillation, and mixed-precision training. A quantitative evaluation pipeline was also built alongside the model: `generate_transla` runs a trained generator over the HiRISE test set, `prepare_real_rgb.py` converts the real rover test set to a matching RGB format, and `compute_fid.py` computes the Fréchet Inception Distance between the two, using `clean-fid`’s InceptionV3 feature extractor with a locally reimplemented Fréchet-distance formula (the installed `clean-fid` version’s own implementation is incompatible with the environment’s current `scipy`).

4.3 A Data-Quality Bug and the Real-Resolution Pipeline Rebuild

An early training run (25 epochs, on the initial corpus) completed and was evaluated at FID 171.8. Before proceeding further, an audit of the actual HiRISE imagery being used revealed a data-quality bug: the corpus was built from HiRISE *browse* JPEGs (`PDS/EXTRAS/RDR`, effectively ~ 3 m/pixel after JPEG compression and downsampling), not the true 25 cm/pixel radiometric data product (`RDR`, `PDS/RDR`) the methodology specifies. Continuing to train and evaluate on browse-quality imagery would silently misrepresent the resolution the dissertation’s Stage 2 baseline is meant to operate at, so the corpus was rebuilt from genuine full-resolution products rather than continuing to iterate on the bugged data.

The corrected pipeline (`hirise_fullres.py`) differs from the original browse-image downloader in several respects, driven by two hard constraints: the training machine’s home directory is NFS-backed and quota-limited (a few gigabytes free), and each real RDR product is a single ~ 568 MB 16-bit JP2 file, so at most one can be on disk at a time.

- **Real RDR URL construction.** Only `_RED.JP2` entries (single-band, full-resolution, non-colour products) are used, filtered from the HiRISE footprint index; this mirrors the existing browse-image filter but points at the true RDR product tree rather than the browse-image tree.

- **Verified download-then-delete lifecycle.** Each observation’s JP2 is downloaded, its size checked exactly against the HTTP `Content-Length` header (a truncated download is deleted and logged as failed rather than silently processed), used to extract patches, and then deleted — via a `try/finally` that runs on both success and failure — before the next observation begins, so disk usage never exceeds one source file at a time.
- **Randomised, entropy-filtered patch sampling.** Patch positions are sampled randomly (with a fixed seed for reproducibility) across each image’s full extent rather than filled sequentially from one corner, then filtered by the same entropy threshold as the original pipeline, up to a target of 5,000 qualifying patches per observation (accepting fewer, rather than lowering the entropy threshold, if an observation does not yield enough).
- **Shared split logic.** The train/val/test splitting logic already used by the browse-image pipeline was extracted into a reusable function (`split_and_move`) so both pipelines split patches identically rather than duplicating that logic.
- **CLI orchestration and corpus replacement.** A command-line entry point selects real-RED-JP2 observations within a target region (Oxia Planum) from the HiRISE footprint index, processes them one at a time, and — once all observations are processed — replaces the existing browse-resolution HiRISE corpus with the new one, writing a manifest recording each observation’s status and patch yield.

Running this pipeline against ten targeted Oxia Planum observations, seven yielded usable RED JP2 products (the remaining three were not present as real RED JP2 entries in the footprint index for the requested region), all seven processed successfully with no download or read failures, producing 30,229 patches (24,183/3,022/3,024 train/val/test) against a target of $\sim 50,000$ — a smaller corpus than targeted, but one built entirely from genuine 25 cm/pixel radiometric data rather than compressed, resampled browse imagery. This corpus, not the original browse-resolution one, is what the training run reported in Chapter 5 uses.

4.4 Summary

The infrastructure, data pipelines, model, and evaluation tooling needed to train and assess a Stage 2 CycleGAN baseline are now in place and have been exercised end-to-end, including recovering from a data-quality bug discovered partway through rather than continuing to build on it.

Chapter 5 reports the results of training this baseline on the corrected corpus, and diagnoses why its FID falls well short of the target despite the corpus now being at the intended resolution.

5 CYCLEGAN BASELINE: TRAINING, EVALUATION AND DIAGNOSIS

This chapter reports the results of training the CycleGAN baseline described in Chapter 4 on the corrected, full-resolution HiRISE corpus, evaluates it quantitatively and visually, and presents a specific, evidence-based diagnosis of why it currently falls well short of the target FID — rather than a general appeal to GAN training instability.

5.1 Training Run

A 50-epoch run (batch size 4, learning rate 2×10^{-4} constant for the first 25 epochs then linearly decayed to zero, $\lambda_{\text{cyc}} = 10$, $\lambda_{\text{idt}} = 5$) was carried out on the A4000, taking approximately 37 minutes per epoch and ~ 31 hours in total. Figure 5.1 shows the full loss history. The generator and cycle-consistency losses converge smoothly throughout — on their own, they would read as a healthy training run. The discriminator losses tell a different story: D_H (the discriminator judging HiRISE realism) settles into a stable equilibrium around 0.03 from epoch 10 onward, while D_R (the discriminator judging rover-image realism) starts *higher* than D_H at epoch 1 (0.162 vs. 0.095), crosses below it by epoch 3, and then decays smoothly and monotonically for the rest of training, reaching 3.99×10^{-6} by epoch 50 (Table 5.1). This divergence is visible well before the learning-rate decay begins at epoch 26, so it is not an artefact of the training schedule.

CycleGAN training — full-res HiRISE corpus, epoch_050 checkpoint

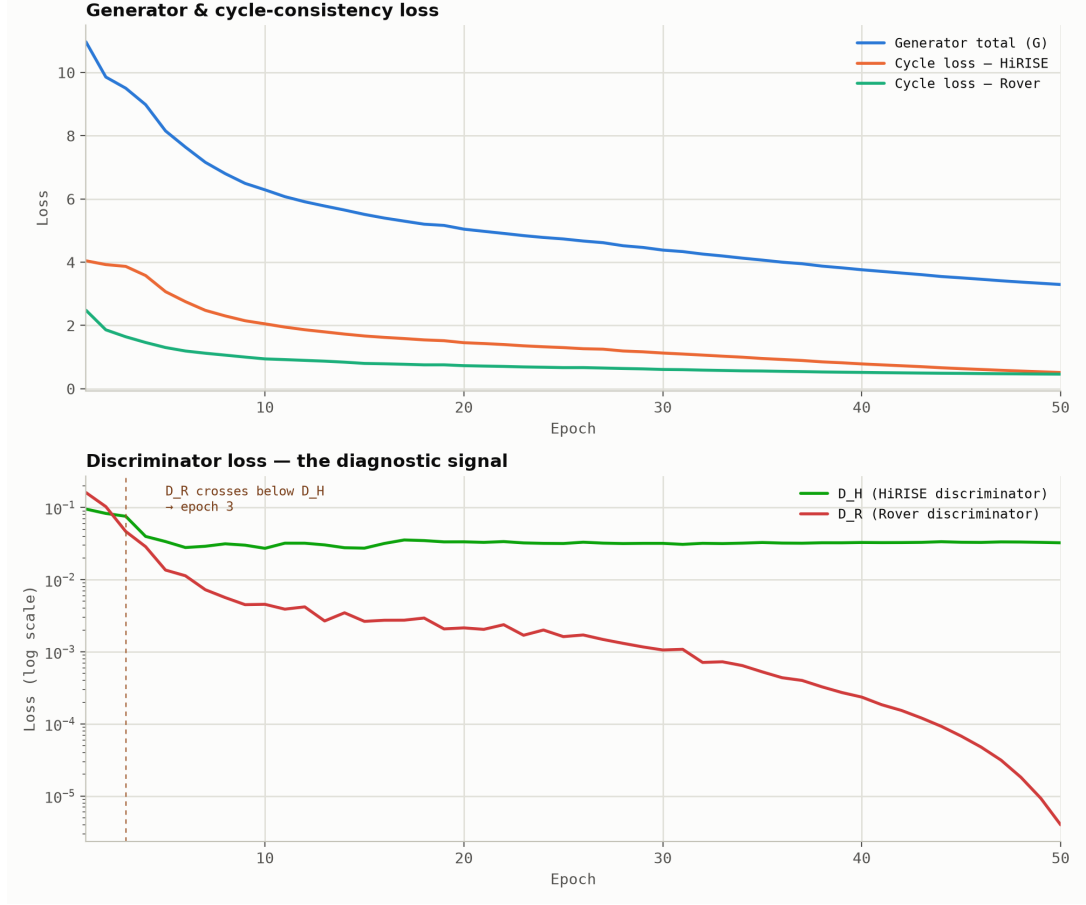


Figure 5.1: Generator/cycle-consistency loss (top, linear scale) and discriminator loss (bottom, log scale) over 50 epochs of training on the corrected full-resolution HiRISE corpus. D_R collapses from epoch 3 onward while D_H remains stable — the diagnostic signal discussed in Section 5.4.

Table 5.1: Discriminator loss at representative epochs, illustrating the D_H/D_R divergence.

Epoch	D_H	D_R
1	0.0954	0.1617
3	0.0758	0.0467
10	0.0272	0.0046
25	0.0318	0.0016
50	0.0325	0.0000040

5.2 Quantitative Evaluation

Table 5.2 reports FID computed between generated and real rover-domain images (Chapter 4’s evaluation pipeline), for both this run and the earlier run on the uncorrected browse-resolution corpus (Section 4.3). Fixing the corpus resolution made FID *worse*, not better — a counter-intuitive result that Section 5.4 explains.

Table 5.2: FID results against the methodology’s target (≤ 55 , Chapter 3).

Run	Checkpoint	Corpus	FID
Prior run	epoch_025	Browse-resolution (~ 3 m/px, buggy)	171.8
This run	epoch_050	Real RDR (25 cm/px, corrected)	269.9
Target (Chapter 3)	—	—	≤ 55

5.3 Visual Inspection of Sample Translations

Sample translation grids saved during training (each showing real HiRISE $\rightarrow$ translated-to-rover $\rightarrow$ reconstructed-HiRISE on top, and real rover $\rightarrow$ translated-to-HiRISE $\rightarrow$ reconstructed-rover below) confirm the same pattern at every epoch inspected, from epoch 1 through the final epoch 50 checkpoint (Figures 5.2–5.4). The HiRISE-to-rover translation stays a grainy, high-frequency echo of the input with no rover-ground structure; the rover-to-HiRISE translation stays a washed-out, low-contrast ghost of the input rather than adopting HiRISE’s characteristic appearance. Both reconstructions (cycle-consistency columns) match their originals closely at every epoch — but, as Section 5.4 argues, that is a symptom of the underlying problem rather than evidence against it.

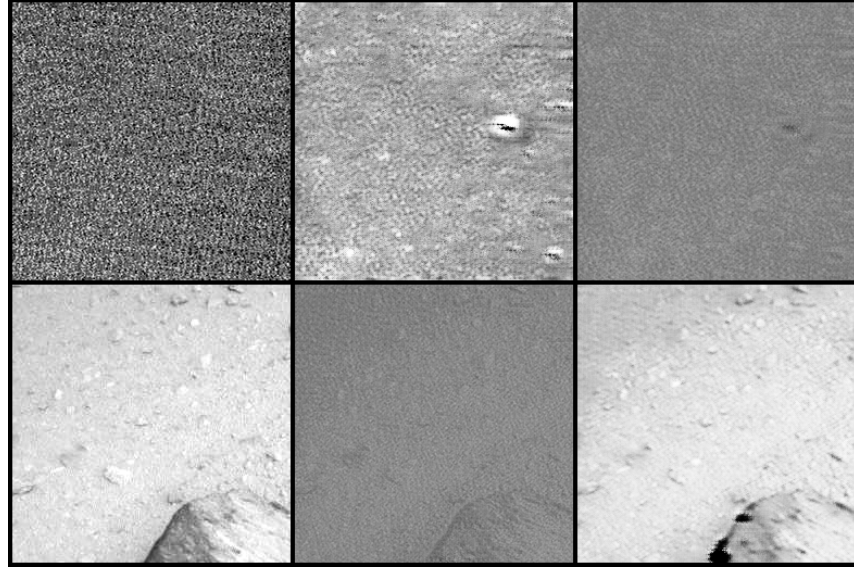


Figure 5.2: Sample translation grid, epoch 1. Real HiRISE input (top-left) already shows the characteristic speckle of genuine 25 cm/pixel radiometric data.

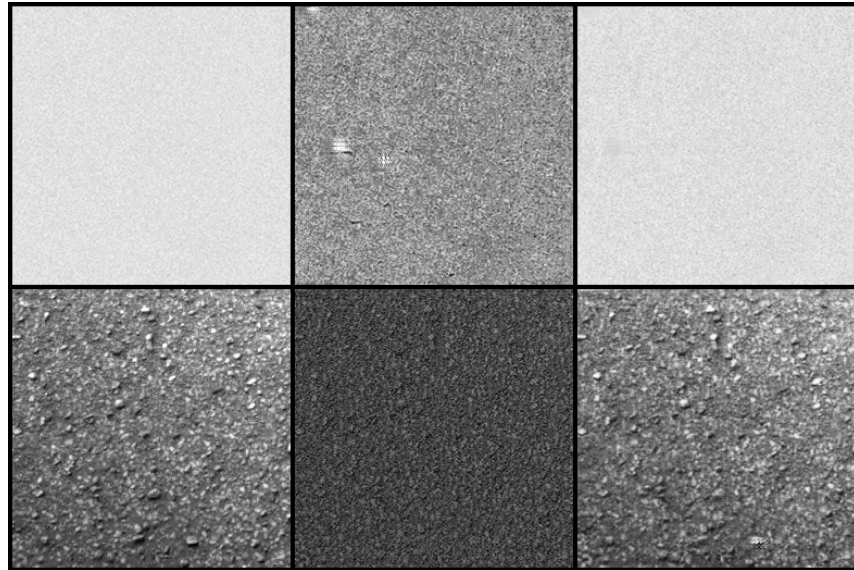


Figure 5.3: Sample translation grid, epoch 30. HiRISE-to-rover output (top-middle) is high-frequency static with no rover-ground structure; D_R is already near collapse at this point (loss ≈ 0.0011).

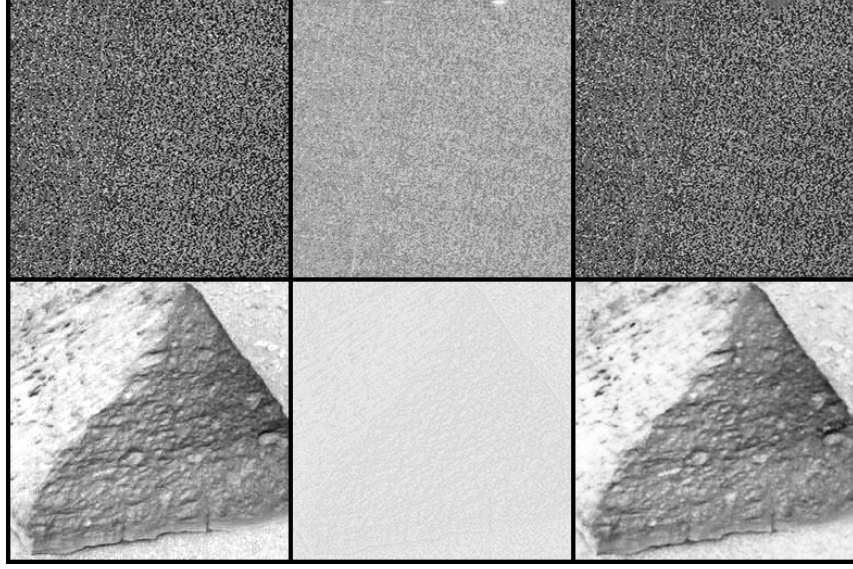


Figure 5.4: Sample translation grid, final epoch 50 checkpoint. The same pattern persists: HiRISE-to-rover stays noise-like, rover-to-HiRISE stays a washed-out echo of the input.

5.4 Diagnosis

The evidence in Sections 4.1–4.3 converges on a specific, mechanistic explanation rather than a generic “GANs are unstable” account.

D_R judges whether a given image is a genuine rover photograph; it is the discriminator that tells G_{H2R} (the HiRISE-to-rover generator) whether its output is convincing. The corrected HiRISE corpus (Chapter 4) is genuinely grainy — visible sensor speckle from real 16-bit radiometric data, stretched to 8-bit — while real rover NAVCAM photographs are comparatively smooth, well-exposed camera images. That is a low-level *noise-statistics* gap layered on top of the intended *style* gap the network is meant to learn to bridge, and it is a far easier signal for a convolutional discriminator to key on than genuine cross-domain semantics (crater texture versus rock/dust texture): D_R can reject G_{H2R} ’s output purely because it still carries HiRISE-style grain, without ever needing to learn what a real rover photograph actually looks like. Once D_R locks onto that shortcut — by epoch 3, per Table 5.1 — it wins for the rest of training essentially for free, so G_{H2R} receives a real but semantically uninformative gradient, consistent with its output staying a grainy echo of the input at every epoch inspected (Section 5.3). The reverse direction, G_{R2H} turning a smooth rover image grainy, is the easier direction for a generator to fake at least superficially, which is consistent with D_H remaining in a healthier, contested equilibrium throughout.

This account also explains why correcting the corpus resolution made FID worse rather than better (Table 5.2): the earlier, browse-resolution corpus was JPEG-compressed and resampled, and therefore smoother and closer to rover-image smoothness, so it did not hand D_R this shortcut as blatantly. Fixing the data-quality bug was still the correct thing to do — training and evaluating against genuinely wrong-resolution imagery would have made every subsequent result meaningless — but it surfaced a training-dynamics problem that the smoother, wrong corpus had been partially masking.

5.5 Candidate Fixes

Three candidate fixes follow directly from the diagnosis, in rough order of expected leverage versus implementation effort:

1. **Close the noise-statistics gap directly**, either by applying mild denoising to real HiRISE patches or by adding matched synthetic sensor grain to the rover domain, so that D_R can no longer separate real from generated rover images on texture statistics alone and is forced to engage with genuine content.
2. **Weaken D_R specifically** relative to the generators — a lower learning rate or a reduced update frequency for D_R alone — which treats the symptom directly and is the cheapest to try.
3. **Reduce λ_{cyc} below 10**, so that cycle-consistency stops rewarding a near-identity mapping and G_{H2R} is less anchored to reproducing input structure.

Resolving this diagnosis is the immediate next priority, ahead of building the physics-constrained UNSB architecture Chapter 3 specifies, since a well-understood, correctly-balanced CycleGAN baseline is a more useful point of comparison for that later work than one left in its current, diagnosed-but-unresolved state.

BIBLIOGRAPHY

- [1] Aithal, S. K. , Maini, P. , Lipton, Z. C. , and Kolter, J. Z. . Understanding hallucinations in diffusion models through mode interpolation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.
- [2] Cao, J. , Huang, R. , Ye, Z. , Xu, Y. , and Tong, X. . Generative 3D reconstruction of martian surfaces using monocular images. In *ISPRS Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, volume XLVIII-3-2024, pages 51–56, 2024. doi: 10.5194/isprs-archives-XLVIII-3-2024-51-2024.
- [3] Catalano, G. L. and Soccini, A. M. . Inpainting the red planet: Diffusion models for the reconstruction of martian environments in virtual reality. *arXiv preprint arXiv:2510.14765*, 2025.
- [4] Almeida, M. , de, Brydon, G. , Persaud, D. M. , Williamson, J. H. , and Henderson, P. . Neural 3D reconstruction of planetary surfaces from descent-phase wide-angle imagery. *arXiv preprint arXiv:2604.13235*, 2025.
- [5] Giusti, L. , Garcia, J. , Cozine, S. , Suen, D. , Nguyen, C. , and Alimo, R. . MaRF: Representing mars as neural radiance fields. *arXiv preprint arXiv:2212.01672*, 2022.
- [6] Hapke, B. . *Theory of Reflectance and Emittance Spectroscopy*. Cambridge University Press, Cambridge, UK, 2nd edition, 2012. doi: 10.1017/CBO9781139025683.
- [7] Kerbl, B. , Kopanas, G. , Leimkühler, T. , and Drettakis, G. . 3D Gaussian splatting for real-time radiance field rendering. In *ACM SIGGRAPH 2023 Conference Proceedings*. ACM, 2023. doi: 10.1145/3588432.3591563.
- [8] Khanna, S. , Liu, P. , Zhou, L. , Meng, C. , Rombach, R. , Burke, M. , Lobell, D. B. , and Ermon, S. . DiffusionSat: A generative foundation model for satellite imagery. In *International Conference on Learning Representations (ICLR)*, 2024.
- [9] Kim, B. , Kwon, G. , Kim, K. , and Ye, J. C. . Unpaired image-to-image translation via neural Schrödinger bridge. In *International Conference on Learning Representations (ICLR)*, 2024.

- [10] Li, L. , Fan, Z. , Cong, W. , Liu, X. , Yin, Y. , Foutter, M. , Pan, P. , You, C. , Wang, Y. , Wang, Z. , Zhao, Y. , Pavone, M. , and Wei, Y. . Martian world model: Controllable video synthesis with physically accurate 3D reconstructions. *arXiv preprint arXiv:2507.07978*, 2025.
- [11] Lu, C. and Su, C. . Super resolution reconstruction of mars thermal infrared remote sensing images integrating multi-source data. *Remote Sensing*, 17(13):2115, 2025. doi: 10.3390/rs17132115.
- [12] Martínez-Petersen, A. , Gerdes, L. , Rodríguez-Martínez, D. , and Pulgar, C. J. , Pérez-del. High-fidelity 3D reconstruction for planetary exploration. *arXiv preprint arXiv:2602.13909*, 2025.
- [13] Osadnik, R. , Gómez, P. , Bohacek, E. , and Bahia, R. . MCTED: A machine-learning-ready dataset for digital elevation model generation from mars imagery. *arXiv preprint arXiv:2509.08027*, 2025.
- [14] Paul, A. , Deevi, S. A. , Chaudhary, A. , Lagisetty, R. K. , Anoop G. L., and Ganesh Kumar R. Neural radiance fields in space applications: A comprehensive review. *International Journal of Advanced Research*, 13(3):525–559, 2025. doi: 10.21474/IJAR01/20596.
- [15] Rathkopf, C. . From hallucination to reliability: Generative modeling and the structure of scientific inference. *arXiv preprint arXiv:2504.08526*, 2025.
- [16] Serrano, N. , McGuire, P. , Mayer, D. , Huertas, A. , and Arvidson, R. . Predicting HiRISE-equivalent rock density on mars using CTX image features. In *AIAA SPACE 2013 Conference and Exposition*. American Institute of Aeronautics and Astronautics, 2013.
- [17] Tao, Y. , Conway, S. J. , Muller, J.-P. , Putri, A. R. D. , Thomas, N. , and Cremonese, G. . Single image super-resolution restoration of TGO CaSSIS colour images: Demonstration with perseverance rover landing site and mars science targets. *Remote Sensing*, 13(9):1777, 2021. doi: 10.3390/rs13091777.
- [18] Tao, Y. , Muller, J.-P. , Xiong, S. , and Conway, S. J. . MADNet 2.0: Pixel-scale topography retrieval from single-view orbital imagery of mars using deep learning. *Remote Sensing*, 13(21):4220, 2021. doi: 10.3390/rs13214220.
- [19] Wu, F. , Jiang, X. , Fu, T. , Fu, Y. , Xu, D. , and Zhao, C. . RSTSRN: Recursive swin transformer super-resolution network for mars images. *Applied Sciences*, 14(20):9286, 2024. doi: 10.3390/app14209286.

- [20] Xiao, B. , Wang, P. , He, Q. , and Dong, M. . Deterministic image-to-image translation via denoising Brownian bridge models with dual approximators. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [21] Zhu, C. , Liu, Y. , Huang, S. , and Wang, F. . Taming a diffusion model to revitalize remote sensing image super-resolution. *Remote Sensing*, 17(8):1348, 2025. doi: 10.3390/rs17081348.
- [22] Zou, Z. , Zhou, Y. , Li, C. , and Dong, Z. . PFMGAN: A generative adversarial network with physics fusion for DTM generation from martian monocular images. *Remote Sensing*, 18(9): 1354, 2026. doi: 10.3390/rs18091354.